REINFORCEMENT LEARNING PROJECT

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1. Description of the setting

The agent starts with $W_0 = 1,000,000$ in wealth in year $t = 0$, needing to withdraw $\omega := \beta \cdot W_0 = 40,000$. At the very beginning of the year t, the agent

(i) has withdrawn a (constant) real amount ω at the very end of last year $t-1$, to feed consumption during year t;
(ii) must decide how to allocate the remaining, post-withdrawal end-of-$(t-1)$ real wealth W_{t-1} among the available assets (stocks, bonds, real estate, and bills aka short-term bonds), by choosing weights a_t in the $(A-1)$ simplex $P_A \subset \mathbb{R}^A$ where A is the number of asset classes.

To make this decision, the agent has access to the following information:

(i) the "observation" o_{t-1} of all important NOMINAL returns that transpired year-on-year during period $t-1$. In particular,

$$o_{t-1} = [r^{eq}_{t-1}, r^{bd}_{t-1}, r^{re}_{t-1}, r^{bill}_{t-1}; \quad r^{gdp}_{t-1}; r^{cpi}_{t-1}].$$

where first are the nominal returns on all asset classes over the year $t-1$, followed by nominal returns over on key macro variables, such as gdp. Note that the nominal return on cpi, r^{cpi}_{t-1}, is by definition the inflation rate over year $t-1$. We adopt the convention that inflation is always the last entry of o_t: $r^{cpi}_t = o_t[-1]$.

We convert this nominal observation o_t into a real observation o^R_t by subtracting off r^{cpi}_t:

$$o^R_{t-1} = [r^{eq}_{t-1} - r^{cpi}_{t-1}, r^{bd}_{t-1} - r^{cpi}_{t-1}, r^{re}_{t-1} - r^{cpi}_{t-1}, r^{bill}_{t-1} - r^{cpi}_{t-1}; \quad r^{gdp}_{t-1} - r^{cpi}_{t-1}]$$

where first are the real (i.e. inflation-adjusted) returns on all asset classes over the year $t-1$, followed by real returns over on key macro variables, such as gdp. Since it is now missing the last dimension, $dim\, o^R_t = dim\, o_t - 1$.

(ii) real wealth W_{t-1} at the end of last year.

We bundle real returns and the real wealth together into a "state" $s_{t-1} := o^R_{t-1} \oplus W_{t-1}$.

Thus, observations o_t happen to the investor, and the investor takes "actions" in response, in the form of a_t and W_t. We will reserve the word "action" for a_t only, as in our model, W_t will be completely formulaic, derived from all other quantities. We explain that now.

1.1. **The withdrawal process.**

The "investor in retirement" that we wish to model is one who maintains a steady lifestyle every year of retirement. That means maintaining a steady stream of real consumption every year. We peg this ideal real withdrawal as $\omega := \beta W_0$, where β is a fraction of starting wealth, like $\beta = 0.04$.

So the agent acts as if: he withdraws ω real dollars at the end of year $t-1$, spends it immediately on ω utils of goods to consume during year t, then and (perhaps progressively, perhaps all at once) consumes those purchased goods over the course of the ensuing year t.

Critique: it is a bit unrealistic, since even if people withdraw only once during the year, they do not spend it all immediately for goods to consume over the coming year. But I think our simplifying single-withdrawal-per-year assumption is relatively harmless.

1.2. **Failure condition: financial ruin.**

For the withdrawal ω to go through successfully, the remaining real wealth W_t must be $\geqslant 0$. This becomes part of a "done" condition: if $W_t = 0$ at the end of a year t, then the agent reaches financial ruin: they withdraw as much as they can, set $done_t = 1$ for "yes", and drop out of the investment world: they have gone destitute, and will be unable to collect profits from all coming returns, including o_t.

1.3. **Actions are allocations.** If the withdrawal at the end of year $t-1$ (for real consumption during the coming year t) is successful, what remains is to determine the new allocation a_t on the remaining real wealth W_{t-1}. Thus, allocation is the entire action.

Given the NEW allocation choice a_t, a NEW observation o_t of real returns at the end of the year t increases the invested real wealth W_{t-1} by

$$\langle o_t^A, a_t \rangle W_{t-1}.$$

Withdrawing ω_t for the coming year $t+1$ then gives a final end-of-t real wealth of

$$W_t = (1 + \langle o_t^A, a_t \rangle) W_{t-1} - \omega_t$$

that gets packaged into the state s_t as the final component: the new state $s_t = o_t \oplus W_t$ is in part a function of the old state s_{t-1} (via W_{t-1}) and the new action a_t. Note crucially that, while W_t depends on W_{t-1}, ω_t does NOT depend on ω_{t-1}: it only adjusts the real, fixed ω in anticipation of the coming year.

SUMMARY: order of events is:

$$\ldots, s_{t-1}, r_{t-1}, done_{t-1}, a_t, s_t, r_t, done_t, \ldots$$

where r_t is the "reward", to be discussed next, acquired at the very end

2. Reward functions

If having observed the state s_{t-1} the agent takes action a_t, and the environment then transitions to state s_t, the agent gets a reward $r_t := r(s_t, a_t, s_{t-1})$ THAT IT FEELS OVER THIS TIME PERIOD, through its end. Thus, rewards are calculated at the very end of t. The form of this reward function will shape the agent's behavior:

(i) suppose the investor will live until age T; then the "characteristic" reward

$$r_c(s_t, a_t, s_{t-1}) := \begin{cases} 1 & \text{if } W_{t-1} \geqslant 0, \\ -k(T-t) & \text{otherwise} \end{cases}$$

just keeps track of whether the investor managed to live the whole year t on full, inflation-adjusted consumption: in other words, the withdrawal ω_t was successful. The moment a withdrawal is not successful, the agent incurs a penalty proportional to the number of years of retirement that they have left (here $k > 0$, and I'm thinking $k \in \{1, 5, 10\}$): effectively, they feel a penalty of k for each remaining year, including t, they have to live without money. The hope: **an agent with this reward structure tries to minimize probability of financial ruin.** Note that this reward is a function only of the previous state s_{t-1}.

(ii) the "wealth" reward

$$r_w(s_t, a_t, s_{t-1}) := W_t - W_{t-1}$$

is simply the change in real wealth. Note this quantity only has meaning because both W_t, W_{t-1} are real wealths, measured in utils; if they were nominal, a fair comparison could only be made by deflating W_t by the realized inflation rate over the year t before taking the difference.

The hope: **an agent with this reward structure tries to maximize the bequest/end-of-life wealth W_T.** This reward is a function of all three inputs.

3. General formulation of RL problem

Given a policy π and an initial state $s_0 \sim \rho$, we can sample the policy $a_t \sim \pi(-|s_{t-1})$ and the environment $s_t \sim P(-|a_t, s_{t-1})$ alternately in order to produce trajectories

$$\tau = (s_0, a_1, s_1, a_2, s_2, \ldots).$$

The set of all possible trajectories thus follows a distribution fully determined by the triple ρ, P, π; while ρ, P are fixtures of the environment, π is adjustable by the agent. The expected value of following a policy π is therefore

$$J(\pi) := \mathbb{E}_{\tau \sim \pi, P}\left[\sum_{t=1}^{\infty} r_t(\tau)\right]$$

where $r_t(\tau) := r(s_t, a_t, s_{t-1})$ is the reward obtained on trajectory τ at time t. The goal of reinforcement learning is to find a π that maximizes J:

$$\pi^* := \text{argmax}_\pi J(\pi).$$

As written, $J(\pi)$ is impossible to calculate precisely. But, as an expectation, it is possible to estimate.

3.1. **Estimates of J.** There are two common way to estimate the policy objective J:

(i) Estimate $J(\pi)$ by sampling a number of full trajectories $\tau \in T$:

$$J(\pi) \approx J_T(\pi) := \frac{1}{|T|} \sum_{\tau \in T} \left[\sum_{t=1}^{\infty} r_t(\tau) \right].$$

(ii) Estimate $J(\pi)$ by further estimating $J_T(\pi)$, by sampling a collection D of individual transitions (s_{t-1}, a_t, s_t) from the set of all transitions in T:

$$J_T(\pi) \approx J_D(\pi) := \frac{1}{|D|} \sum_{(s_{t-1}, a_t, s_t)} r(s_{t-1}, a_t, s_t).$$

Note that for either estimate to be "accurate" (specifically: unbiased), the trajectories T must be produced by π itself. For either estimate to have low variance, $|T|$ and $|D|$ must be large. Using a different policy π' to sample trajectories or transitions will degrade the estimate of the objective for π by the size of the discrepancy between π, π'. The difference between the current policy π and the policy π' from which samples are taken is called **distribution shift**.

Any procedure of trying to discover π^* will feature a long line of policies that iteratively innovate on each other:

$$\pi_1, \pi_2, \ldots \pi_n, \pi_{n+1}, \ldots, \pi^*?$$

An innovation $\pi_n \mapsto \pi_{n+1}$ makes use of trajectory data T_n gathered under policy π_n:

$$\pi_{n+1} = \pi_n + T_n$$

In light of estimating the policy objectives $J(\pi_n)$, this gradual accumulation of trajectory information $T_1, T_2, \ldots$ under different historical policies produces a trade-off:

(i) On the one hand, the estimate of $J(\pi_n)$ that is **least biased** is the one that only uses the trajectories T_n gathered under π_n itself.

(ii) On the other hand, the estimate of $J(\pi_n)$ that has **least variance** is the one that uses the most data. One idea: using the combined trajectories $T = T_n \cup_{i=1}^{k} T_{n_i}$ from k additional policies might reasonably give a lower variance for the estimate than T_n from π_n alone. To make this idea most reasonable, we should pick the k policies to be the ones we believe are closest to π_n.The most natural choices for such policies are those from the past k innovations, $\{\pi_{n-k+i}\}_{i=0}^{k-1}$.

While either method of estimation can be paired with either end of the data utilization trade-off spectrum, estimate (i) most naturally pairs with the low bias approach (i), while estimate (ii) most directly accommodates the low variance approach (ii). This effectively splits reinforcement learning into two modes:

(i) **on-policy**, using just the most recent policy π to estimate $J_T(\pi)$ by unrolling trajectories T under π alone; this is the least biased. This uses each unique transition **exactly once**. To lower variance, an on-policy approach must increase the number of trajectories T collected under π.

(ii) **off-policy**, using policies $\pi' \neq \pi$ (hopefully not too different from π) to gather and store transitions in a **replay buffer** of transitions $\mathcal{D} = \{(s_{t-1}, a_t, s_t)\}$ from which reasonably large samples $D \subset \mathcal{D}$ are used to produce an estimate of $J_D(\pi)$; this has the promise of having lower variance than an on-policy method, and of being more **data-efficient** in the sense of making repeated use of transitions, gathered from past policies $\pi' \neq \pi$. To reduce bias, an off-policy method must maintain a replay buffer $\mathcal{D}$ that is reasonably up-to-date: its transitions must come from pursuing policies π' that are not too old. This is achieved by

1. capping the size of $\mathcal{D}$ to some $|\mathcal{D}| \leqslant M$. Think of the cap as the product $M = k \cdot n \cdot T$ where k is the largest number of innovations that can separate a policy π' from the current π and for that π' to still have samples that are represented in the current $\mathcal{D}$, n is the number of full

trajectories unrolled and committed to $\mathcal{D}$ for each policy, and $T := \mathbb{E}|\tau|$ is the average length of a trajectory under the last k policies.

2. overwriting oldest transitions with newest. We can do this easily by creating a list of size M and a counter c recording the order of each transition, and committing transition c to index c mod M in the buffer.

3.2. **Value functions.** For the eventual purpose of implementation, we will re-formulate things in terms of value functions V instead of the functions J above.

Given a state s_t, the value function $V^\pi(s_t)$ is meant to communicate the expected future rewards, discounted to the BEGINNING of time t (end of time s_{t-1}), that following the policy π from state s_{t-1} will produce:

$$\begin{aligned} V^\pi(s_{t-1}) &= \mathbb{E}_{\substack{a_t \sim \pi(-|s_{t-1}) \\ s_t \sim P(-|s_{t-1}, a_t)}} \left[r(s_t, a_t, s_{t-1}) + \gamma_t V^\pi(s_t) \right] \\ &= \mathbb{E}_{a_t \sim \pi(-|s_{t-1})} [Q^\pi(s_{t-1}, a_t)] \end{aligned}$$

where

$$Q^\pi(s_{t-1}, a_t) = \mathbb{E}_{s_t \sim P(-|s_{t-1}, a_t)} [r(s_t, a_t, s_{t-1}) + \gamma_t V^\pi(s_t)]$$

is the expected benefit of selecting action a_t having seen s_{t-1}, and then running the policy π thereafter. Thus the relation to J-functions above is

$$J(\pi) = \mathbb{E}_{s_0 \sim \rho} V^\pi(s_0)$$

Remark 3.1. For us, there is/might be a unitless (and inflation-independent! since we reformulatd everything in terms of real returns and wealth) "discounting" factor $\gamma_t = \gamma(s_t, a_t, s_{t-1})$ that reflects the reward-equivalent at time t of one unit of reward guaranteed at time $t+1$. This γ_t might in principle depend on states and actions, and will also differ based on the reward function chosen:

(i) if the reward is r_c, then reward and value are unitless. Then γ_t **???interpretation?** maybe set it to $\gamma_t = 1$?

(ii) if the reward is r_w, then reward and value V^π have units of real wealth. Since the units are real wealth, the natural choice

$$\gamma_t(s_t, a_t, s_{t-1}) = (1 + o_t[-1])^{-1},$$

the inverse of inflation for year t, is INCORRECT: inflation is already accounted for in the values W_t. Instead, γ_t includes investor details and preferences: current real wealth, current risk aversion at the level of real wealth, etc.

4. Details of this project [HIGHLY EXPERIMENTAL SECTION].

We elect to implement a "vanilla" (not "Soft") Actor-Critic model, which is **off-policy**. We will need to describe

(i) the buffer
(ii) the policy training
(iii) (it will turn out) the Q-function training

4.1. **Buffer $\mathcal{D}$.** Since we pursue an off-policy method, we will therefore need to build and maintain a replay buffer $\mathcal{D}$, and decide on its size. Additionally, since the buffer starts empty, we will need to decide how long to wait before starting to sample from $\mathcal{D}$.

An "innovation" for us will be a gradient update. To make a gradient update, we will sample $D := |D|$ transitions from $\mathcal{D}$. Recalling that

$$M = k \cdot n \cdot T,$$

the bigger the n, the higher the proportion in $\mathcal{D}$ of transitions from the current policy, and the more current $\mathcal{D}$ generally is. Similarly, the smaller the k, the more current the $\mathcal{D}$.

4.1.1. *Replay buffer size.* In the investing problem, trajectory lengths are capped, on average, by the life expectancy of a person past retirement: this is around 20 years (time steps). This is also independent of the policy: if a policy causes an investor to go into financial ruin prior to death, the investor continues to live, but accrues a penalty for each remaining year. Thus, $T = 20$.

Our dataset consists of about 2700 unique observation transitions (o_{t-1}, o_t): 18 countries, each with 150 years of return history, for a total of $18 \times 150 = 2700$ observation transitions. This means that 135 typical trajectories will produce the same number of transitions as there are total. Let us assert (reason???) it sensible that

$$M \leqslant 2700$$

and thus that

$$k \cdot n \leqslant 135.$$

Recall that if the goal is for $\mathcal{D}$ to be current, then we wish to have n big and k small. Let us take

$$k = 10, \quad n = 10,$$

giving a buffer of maximum size

$$M = k \cdot n \cdot T = 2000.$$

It remains to pin down the number of samples D from $\mathcal{D}$ needed to perform a policy update. Whatever D is, it should be

$$n \cdot T << D \leqslant M,$$

since if it were too small then we should just be on-policy and use the $n \cdot T$ transitions gathered under π. The power of off-policy learning comes from re-using samples from prior-but-not-too-distant policies.

Let us assert that $D = \ell \cdot n \cdot T$ for some **"lookback" time** ℓ, $2 \leqslant \ell \leqslant k$, allowing us to pick up as much data as was generated by the past ℓ policies. Perhaps a reasonable choice is $\ell = \frac{k}{2}$—half the size of the buffer.

4.1.2. *Wait time.* Since we require D transitions to compute a gradient step, it makes sense to wait at least until there are D transitions in the buffer. Let us choose to wait exactly this long (acquiring just a bunch of random transitions).

In sum, here is how the buffer works:

(i) Knowing that we have only about 2700 unique environment observation transitions (though many more state transitions), we set
  1 $n = 10$ trajectories/policy (want it big)
  2 $k = 10$ historical policies on record (want it small)
  3 $\ell = \lfloor \frac{1}{2}k \rfloor = 5$ "lookback" time to calculate number of samples D
  4 $D = \ell \cdot n \cdot T = 1000$ number of samples
  5 $M = k \cdot n \cdot T = 2000$ max buffer size.

(ii) initialize buffer $\mathcal{D}$ of size M,

(iii) wait D time steps before beginning gradient steps,

(iv) each gradient step requires sample of D transitions from $\mathcal{D}$, and some yet undiscussed number N of samples from the current policy (see training details below).

4.2. **Training the policy.** In theory, we look for a policy π that maximizes

$$J(\pi) := \mathbb{E}_{s_0 \sim \rho} V^{\pi}(s_0) = \mathbb{E}_{s_0 \sim \rho} \mathbb{E}_{a \sim \pi(-|s_0)}[Q^{\pi}(s_0, a)]$$

where s_0 is a starting state, distributed according to the environment's distribution of starting states ρ. Thus, we look for a π that maximizes the expected discounted value, averaged over starting states.

In practice, we make some compromises for the sake of computability, and change our optimization problem:

(i) (Parametrization) To maximize an objective by gradient descent, we must restrict π to a parametric family $\{\pi_\theta\}_{\theta \in \Theta}$.

(ii) (Decouple Q from π) We cannot calculate $Q^{\pi_\theta}(s,a)$ directly. However, we may estimate it by unrolling trajectories. Additionally, and commonly, we may decouple Q from π by making it a separate parametric function $\{Q_\phi\}_{\phi\in\Phi}$, and optimize Q separately; the hope is that as π_θ and Q_ϕ jointly evolve, they converge to functions $\pi_{\theta*}, Q_{\phi*}$ for which $Q_{\phi*}$ is (or is a decent approximation to) the Q-function of $\pi_{\theta*}$. We will need to make design choices to support that hope.

(iii) (Use two Q networks) To control the value and variance of Q_ϕ, we stay pessimistic: we introduce a second Q-network with its own parameter — so now we have Q_{ϕ_1}, Q_{ϕ_2}, and replace Q_θ in $J(\theta)$ by the minimum of the two Q-values.

(iv) (Use a replay buffer $\mathcal{D}$) Note that $s_0 \sim \rho, a \sim \pi(-|s_0)$ is a state-action transition, which just happens to start with the initial state s_0. To make use of our wealth data, we will replace this by a general transition—s_0 will no longer be distributed according to ρ, but will typically be pulled from the middles of trajectories. To that end, we will maintain a replay buffer $\mathcal{D}$ of trajectories (s_{t-1}, a_t, s_t), obtained by policies from the recent past, and take uniformly random samples $D \subset \mathcal{D}$ from them to approximate expectation values.

Taken together, our refined policy objective on a batch $D = \{(s_{t-1}, a_t, s_t)\}$ of transitions is

$$\begin{aligned} J_D(\theta|\phi_1,\phi_2) &:= \mathbb{E}_{(s_{t-1},a_t,s_t)\sim D}\left[\mathbb{E}_{\tilde{a}_t\sim\pi_\theta(-|s_{t-1})}\left[\min_{i=1,2}\{Q_{\phi_i}(s_{t-1},\tilde{a}_t)\}\right]\right] \\ &= \frac{1}{D}\sum_{(s_{t-1},a_t,s_t)\in D}\frac{1}{N}\sum_{\tilde{a}_t\sim\pi_\theta(-|s_{t-1})}\min_{i=1,2}\{Q_{\phi_i}(s_{t-1},\tilde{a}_t)\} \end{aligned}$$

where $D := |D|$ is the number of transitions sampled, and N is the number of samples $\tilde{a}_t \sim \pi_{\theta(-|s_{t-1})}$ taken of the current policy. The key move to observe here is that we have sampled a transition (s_{t-1}, a_t, s_t) from D, but only to use the first part of it: s_{t-1} serves as a "reasonable state to start in, given the entire history of states experienced in training, up until now." We drop the remaining a_t, s_t from the transition in favor of sampling an action $\tilde{a}_t \in \pi_\theta(-|s_{t-1})$ from the most up-to-date policy π_θ, so that we may differentiate the objective with respect to θ.

Here is the role of maximizing this term, where ϕ_i are frozen: this term samples the current policy π_θ given a starting state s_- drawn from a historical hat. Its meaning rests on two assumptions:

(i) the Q_{ϕ_i}, or at least their minimum, are fair approximations to the Q-function of π_θ, given current values (θ, ϕ_1, ϕ_2);

(ii) the state s_- is drawn from a distribution on visited states that is similar to the distribution determined by π_θ.

In practice, both are very false: the Q_{ϕ_i}, at least initially have some, but little connection to π_θ, and the transition (s_-, a, s) was sampled potentially by a much older distribution $\pi_{\theta_{old}}$ from which π_θ might have shifted significantly. We can mitigate the effect of (ii) by **reducing the size of the replay buffer of all stored transitions, and overwriting the oldest transitions first as new trajectories unroll under new, updated policies.** Hopefully this will keep the transitions in the buffer reasonably close to those that π_θ would have made (though of course they will necessarily lag).

Having made a design choice to support the validity of assumption (ii), we must make further design choices to support the validity of assumption (i). Thus we come to a discussion of how to train the Q_{ϕ_i}.

4.3. **Training the critics.** Begin by recalling the formula for Q^π, and rewrite it in recursive form:

$$\begin{aligned} Q^\pi(s_{t-1},a_t) &= \mathbb{E}_{s_t\sim P(-|s_{t-1},a_t)}[r(s_t,a_t,s_{t-1}) + \gamma_t V^\pi(s_t)] \\ &= \mathbb{E}_{s_t\sim P(-|s_{t-1},a_t)}[r(s_t,a_t,s_{t-1}) + \gamma_t \mathbb{E}_{a_{t+1}\sim\pi(-|s_t)}[Q^\pi(s_t,a_{t+1})]] \\ &= \mathbb{E}_{s_t\sim P(-|s_{t-1},a_t)}\mathbb{E}_{a_{t+1}\sim\pi(-|s_t)}\left[r(s_{t-1},a_t,s_t) + \gamma_t Q^\pi(s_t,a_{t+1})\right] \end{aligned}$$

The last line made use of the reward's independence of the next action a_{t+1}.

Replacing π by π_θ and Q^π by Q_{ϕ_i} gives no longer an equation, but an aspiration:

$$Q_{\phi_i}(s_{t-1},a_t) \stackrel{?}{=} \mathbb{E}_{s_t\sim P(-|s_{t-1},a_t)}\mathbb{E}_{a_{t+1}\sim\pi_\theta(-|s_t)}\left[r(s_{t-1},a_t,s_t) + \gamma_t Q_{\phi_i}(s_t,a_{t+1})\right]$$

The extent to which this equality fails is the extent to which Q_{ϕ_i} fails to be the Q-function of π_θ. A natural idea is to change ϕ_i so that the distance between the the left-hand side and the right-hand side is

minimized, for example by taking the mean squared error over a number of transitions $D = \{(s_{t-1}, a_t, s_t)\}$, and taking N samples of the current policy $a_{t+1} \sim \pi_\theta(-|s_t)$:

$$\widetilde{J}_D(\phi_i|\theta) := \mathbb{E}_{(s_{t-1},a_t,s_t)\sim D}\left[\left(Q_{\phi_i}(s_{t-1}, a_t) - \frac{1}{N}\sum_{a_{t+1}\sim\pi_\theta(-|s_t)}\left[r(s_{t-1}, a_t, s_t) + \gamma_t Q_{\phi_i}(s_t, a_{t+1})\right]\right)^2\right]$$

Minimizing this again assumes that D (sampled from a historical record generated by old policies) is close to what the current policy π_θ would produce. We have already addressed this concern by discussing the limited size and overwriting of the buffer.

In practice, people have found that an additional trick is needed to improve learning: to replace the second appearance of Q_{ϕ_i}, in the form $Q_{\phi_i}(s_t, a_{t+1})$, by a lagged version of Q_{ϕ_i} with frozen parameters, called a **target network** $Q_{\phi_i^{target}}$. Though frozen and thus unchanged during gradient flow, the parameters of the target networks do nonetheless get updated mechanically towards their respective ϕ_i:

$$\phi_i^{target} \mapsto (1-\tau)\phi_i^{target} + \tau\phi_i$$

where $\tau << 1$ is the exponential rate of approach. Additionally, we smuggle in our pessimism again by taking not just the target Q-values, but their minimum. The final objectives for the two critics are therefore

$$J_D(\phi_i|\theta) := \frac{1}{|D|}\sum_{(s_{t-1},a_t,s_t)\in D}\left(Q_{\phi_i}(s_{t-1}, a_t) - y(s_{t-1}, a_t, s_t|\theta)\right)^2$$

where

$$y(s_{t-1}, a_t, s_t|\theta) := r(s_{t-1}, a_t, s_t) + \frac{1}{N}\sum_{a_{t+1}\sim\pi_\theta(-|s_t)}\left[\gamma_t \cdot \min_{i=1,2}\{Q_{\phi_i^{target}}(s_t, a_{t+1})\}\right]$$

is the **target value**, depending on both full transitions from D and N samples from the current (temporarily frozen) policy π_θ.

4.4. **Summary.** *Email address*: daniel_chupin@berkeley.edu